

Analysis

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2026-09-24

TPOT with full graph search space versus ensemble

500 total evals

Version 1

Stats Testing

```
## # A tibble: 26 x 9
##   task_id .y. group1      group2  n1  n2 statistic      p p.signif
##   <int> <chr> <chr>      <chr> <int> <int>    <dbl>    <dbl> <chr>
## 1   2073 auroc ensemble_random_~ indiv~ 15  15    115 6.1 e-4 ***
## 2  146818 auroc ensemble_random_~ indiv~ 21  21    207 1.56e-3 **
## 3  146820 auroc ensemble_random_~ indiv~ 21  21    168. 6.8 e-2 ns
## 4  168350 auroc ensemble_random_~ indiv~ 21  21    230 1.91e-6 ****
## 5  168757 auroc ensemble_random_~ indiv~ 21  21    166 8.22e-2 ns
## 6  168784 auroc ensemble_random_~ indiv~ 19  19    190 3.81e-6 ****
## 7  168911 auroc ensemble_random_~ indiv~ 20  20    174 8.31e-3 **
## 8  190137 auroc ensemble_random_~ indiv~ 20  20    163 2.96e-2 *
## 9  190146 auroc ensemble_random_~ indiv~ 20  20     92 6.48e-1 ns
## 10 190411 auroc ensemble_random_~ indiv~ 20  20    208 5.72e-6 ****
## 11 359955 auroc ensemble_random_~ indiv~ 21  21    194. 6.36e-3 **
## 12 359956 auroc ensemble_random_~ indiv~ 20  20    187 1.21e-3 **
## 13 359958 auroc ensemble_random_~ indiv~ 21  21    205 1 e-3 ***
## 14 359959 auroc ensemble_random_~ indiv~ 21  21    222 3.15e-5 ****
## 15 359960 auroc ensemble_random_~ indiv~ 20  20     51 1.2 e-1 ns
## 16 359962 auroc ensemble_random_~ indiv~ 21  21    189 9.02e-3 **
## 17 359963 auroc ensemble_random_~ indiv~ 21  21    225 1.34e-5 ****
## 18 359964 auroc ensemble_random_~ indiv~ 20  20    181 3.15e-3 **
## 19 359965 auroc ensemble_random_~ indiv~ 21  21    165 8.88e-2 ns
## 20 359968 auroc ensemble_random_~ indiv~ 21  21    165 8.88e-2 ns
## 21 359969 auroc ensemble_random_~ indiv~ 20  20    210 1.91e-6 ****
## 22 359970 auroc ensemble_random_~ indiv~ 18  18    171 7.63e-6 ****
## 23 359971 auroc ensemble_random_~ indiv~ 19  19    167 2.4 e-3 **
## 24 359972 auroc ensemble_random_~ indiv~ 21  21    217 1.05e-4 ***
## 25 359974 auroc ensemble_random_~ indiv~ 17  17    130 9.34e-3 **
## 26 359975 auroc ensemble_random_~ indiv~ 19  19    137 9.55e-2 ns
```

Medians

```
## # A tibble: 26 x 3
##   task_id individual_tpot ensemble_random_100
##   <int>          <dbl>          <dbl>
## 1   2073          0.872          0.883
## 2  146818          0.924          0.939
## 3  146820          0.995          0.997
## 4  168350          0.960          0.965
## 5  168757          0.793          0.800
## 6  168784          0.963          0.970
## 7  168911          0.870          0.870
## 8  190137          0.925          0.933
## 9  190146          0.967          0.966
## 10 190411          0.921          0.924
## 11 359955          0.739          0.752
## 12 359956          0.934          0.937
## 13 359958          0.943          0.952
## 14 359959          0.747          0.754
## 15 359960          1          1
## 16 359962          0.830          0.846
## 17 359963          0.994          0.996
## 18 359964          0.994          0.994
## 19 359965          1.00          1.00
```

##	20	359968	0.918	0.923
##	21	359969	0.846	0.855
##	22	359970	0.919	0.933
##	23	359971	0.997	0.997
##	24	359972	0.989	0.990
##	25	359974	0.879	0.882
##	26	359975	0.994	0.996